

Justify all answers. **Answers without sufficient justification will receive no credit. A numeric answer alone, without explanation and supporting work, is not acceptable.** Submit as a pdf to canvas.

1. (20 points) Use existential and universal quantifiers to express the statement “Every grandparent is a parent of a parent” using the propositional functions $G(g, y)$, which represents “ g is a grandparent of y ,” and $P(x, y)$, which represents “ x is a parent of y .”

Possible Solution: Let the domain of discourse be the set of all people.

The statement “Every grandparent is a parent of a parent” can then be written as follows.

$$\forall g \left((\exists y G(g, y)) \Rightarrow \exists x \exists z (P(g, x) \wedge P(x, z)) \right).$$

2. (20 points) Let A and B be sets. Prove that if $A \subseteq B$, then $A \cap B = A$.

Possible Element Chasing Solution: Assume $A \subseteq B$. We show both inclusions.

$(\subseteq)$ Let $x \in A \cap B$. Then $x \in A$ and $x \in B$. In particular, $x \in A$. Hence $A \cap B \subseteq A$.

$(\supseteq)$ Let $x \in A$. Since $A \subseteq B$, we have $x \in B$. Therefore $x \in A \cap B$. Hence $A \subseteq A \cap B$.

Combining the two inclusions, $A \cap B = A$.

Possible Truth Table Solution: Let x be arbitrary. Introduce propositional variables:

$$p : x \in A, \quad q : x \in B.$$

The hypothesis $A \subseteq B$ is $p \rightarrow q$. We want to show $x \in A \cap B$ iff $x \in A$, i.e., $(p \wedge q) \leftrightarrow p$.

p	q	$p \wedge q$	p
F	F	F	F
F	T	F	F
T	F	F	T
T	T	T	T

Under $p \rightarrow q$, the row $p = T, q = F$ is excluded. On the remaining rows, the columns for $p \wedge q$ and p coincide. Hence $(p \wedge q)$ is equivalent to p under the assumption, so $x \in A \cap B$ iff $x \in A$. Therefore $A \cap B = A$.

3. (20 points) Use mathematical induction to show that

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \cdots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1},$$

whenever n is a positive integer.

Possible Solution: We prove by induction on the positive integer n .

Base case ($n = 1$).

$$\frac{1}{1 \cdot 4} = \frac{1}{4} = \frac{1}{3 \cdot 1 + 1},$$

so the statement holds for $n = 1$.

Inductive step. Assume the identity holds for some integer $k \geq 1$, that is,

$$\sum_{m=1}^k \frac{1}{(3m-2)(3m+1)} = \frac{k}{3k+1}.$$

Then, for $k + 1$, we have:

$$\sum_{m=1}^{k+1} \frac{1}{(3m-2)(3m+1)} = \frac{k}{3k+1} + \frac{1}{(3(k+1)-2)(3(k+1)+1)}.$$

Simplify the last fraction:

$$= \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} = \frac{k(3k+4) + 1}{(3k+1)(3k+4)}.$$

Compute the numerator:

$$k(3k+4) + 1 = 3k^2 + 4k + 1 = (k+1)(3k+1).$$

Thus:

$$\sum_{m=1}^{k+1} \frac{1}{(3m-2)(3m+1)} = \frac{(k+1)(3k+1)}{(3k+1)(3k+4)} = \frac{k+1}{3(k+1)+1},$$

which is precisely the formula with $n = k + 1$.

By induction, the identity holds for all positive integers n .

4. (20 points) Solve the recurrence relation together with the initial conditions:

$$a_n = 4a_{n-1} - 4a_{n-2} + 3^n \quad \text{for } n \geq 2, \quad a_0 = 6, \quad a_1 = 8.$$

Possible Solution: We solve the recurrence relation:

$$a_n = 4a_{n-1} - 4a_{n-2} + 3^n, \quad a_0 = 6, \quad a_1 = 8.$$

Step 1: Solve the Homogeneous Recurrence

Consider the associated homogeneous recurrence:

$$a_n^{(h)} - 4a_{n-1}^{(h)} + 4a_{n-2}^{(h)} = 0.$$

The characteristic equation is:

$$r^2 - 4r + 4 = 0.$$

Factor the characteristic polynomial:

$$(r - 2)^2 = 0.$$

We have a repeated root $r = 2$, so the homogeneous solution is:

$$a_n^{(h)} = (C_1 + C_2 n)2^n.$$

Step 2: Find a Particular Solution

Since the non-homogeneous term is 3^n , we try a particular solution of the form:

$$a_n^{(p)} = A \cdot 3^n.$$

Substitute into the recurrence:

$$A \cdot 3^n = 4A \cdot 3^{n-1} - 4A \cdot 3^{n-2} + 3^n.$$

Divide by 3^{n-2} :

$$9A = 12A - 4A + 9 \implies 9A = 8A + 9 \implies A = 9.$$

Thus, a particular solution is:

$$a_n^{(p)} = 9 \cdot 3^n.$$

Step 3: General Solution

The general solution is the sum of the homogeneous and particular solutions:

$$a_n = (C_1 + C_2 n)2^n + 9 \cdot 3^n.$$

Use the initial conditions $a_0 = 6$, $a_1 = 8$:

For $n = 0$:

$$a_0 = (C_1 + C_2 \cdot 0)2^0 + 9 \cdot 3^0 = C_1 + 9 = 6 \implies C_1 = -3.$$

For $n = 1$:

$$a_1 = (C_1 + C_2 \cdot 1)2^1 + 9 \cdot 3^1 = 2(C_1 + C_2) + 27 = 8.$$

Substitute $C_1 = -3$:

$$2(-3 + C_2) + 27 = 8 \implies -6 + 2C_2 + 27 = 8 \implies 2C_2 = -13 \implies C_2 = -\frac{13}{2}.$$

Step 4: Final Explicit Solution

Thus, the explicit solution is:

$$a_n = \left(-3 - \frac{13}{2}n\right)2^n + 9 \cdot 3^n.$$

5. (20 points) You have n identical balloons to give to three children: Alice, Ben, and Chloe.

- Alice can receive **any nonnegative** number of balloons.
- Ben can receive **at most 2** balloons;
- Chloe must receive an **even** number of balloons;

How many ways are there to distribute the balloons?

Possible Solution: (*Solved in class; see Slide 10 of Lecture 12.*)

Let a_n be the number of ways to distribute n identical balloons among Alice, Ben, and Chloe, subject to:

$$x_A + x_B + x_C = n$$

with constraints:

- x_A (Alice): no constraint (any nonnegative integer);
- x_B (Ben): at most 2 balloons;
- x_C (Chloe): even number.

Generating-function encoding:

- x_A (**no constraint**): $1 + x + x^2 + \cdots = \frac{1}{1-x}$.
- x_B (**at most 2**): $1 + x + x^2$.
- x_C (**even**): $1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2}$.

Thus, the generating function is:

$$A(x) = \frac{1}{1-x} \cdot (1 + x + x^2) \cdot \frac{1}{1-x^2} = \frac{1 + x + x^2}{(1-x)^2(1+x)}.$$

Partial fraction decomposition: Express

$$\frac{1 + x + x^2}{(1-x)^2(1+x)} = \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{1+x}.$$

Solve for coefficients:

$$1 + x + x^2 = A(1-x^2) + B(1+x) + C(1-2x+x^2).$$

Matching coefficients for powers of x :

$$\begin{cases} A + B + C = 1, \\ -2C + B = 1, \\ -A + C = 1. \end{cases}$$

Solve the system to get:

$$A = -\frac{3}{4}, \quad B = \frac{3}{2}, \quad C = \frac{1}{4}.$$

Thus:

$$A(x) = -\frac{3}{4} \frac{1}{1-x} + \frac{3}{2} \frac{1}{(1-x)^2} + \frac{1}{4} \frac{1}{1+x}.$$

Closed-form solution: Using known expansions:

$$\frac{1}{1-x} = \sum_{n \geq 0} x^n, \quad \frac{1}{(1-x)^2} = \sum_{n \geq 0} (n+1)x^n, \quad \frac{1}{1+x} = \sum_{n \geq 0} (-1)^n x^n,$$

the coefficient for x^n is:

$$a_n = -\frac{3}{4}(1) + \frac{3}{2}(n+1) + \frac{1}{4}(-1)^n.$$

Simplifying, we have:

$$a_n = \frac{3}{2}n + \frac{3 + (-1)^n}{4}.$$

6. (20 points) A *bitonic array* $A[1..n]$ is one that strictly increases up to a (unique) peak index p and then strictly decreases; that is, $A[1] < A[2] < \cdots < A[p] > A[p+1] > \cdots > A[n]$ for some $1 \leq p \leq n$. Find a recurrence relation that describes the running time of the following algorithm: Find the peak index in a bitonic array of n numbers by performing a binary-search-style procedure that compares the middle element $A[\lfloor n/2 \rfloor]$ with its right neighbor $A[\lfloor n/2 \rfloor + 1]$ to decide whether to recurse on the left half or the right half, repeating until a single index remains. Then apply the Master Theorem to obtain the big- O estimate for the running time of the algorithm.

Possible Solution: Let $T(n)$ represent the running time to find the peak in a bitonic array of size n using the described binary-search-style algorithm.

- At each step, we compare the middle element with its neighbor, at a constant cost $O(1)$.
- After comparison, we recursively search only in one half of the array, halving the search space.

Thus, we have the recurrence relation:

$$T(n) = T\left(\frac{n}{2}\right) + O(1), \quad T(1) = O(1).$$

Applying the Master Theorem:

We match the recurrence to the form of the Master Theorem:

$$T(n) = aT\left(\frac{n}{b}\right) + cn^d,$$

with parameters:

$$a = 1, \quad b = 2, \quad c > 0, \quad d = 0.$$

We now compare a with b^d :

$$a = 1, \quad b^d = 2^0 = 1.$$

Since $a = b^d$, by case 2 of the Master Theorem, we obtain:

$$T(n) = O(n^d \log n) = O(\log n).$$

Final Complexity:

$$T(n) = O(\log n).$$